KOLKATA THIS WEEKEND

DETAILED TECH STACK & SYSTEM ARCHITECTURE

Frontend • Backend • Data • AI • Search • Maps • Analytics

This document translates the product concept into a practical engineering architecture for a Kolkata-first launch and later city expansion.

FRONTEND

Next.js TypeScript Tailwind CSS PWA

BACKEND

Node.js PostgreSQL Supabase REST APIs

INTELLIGENCE

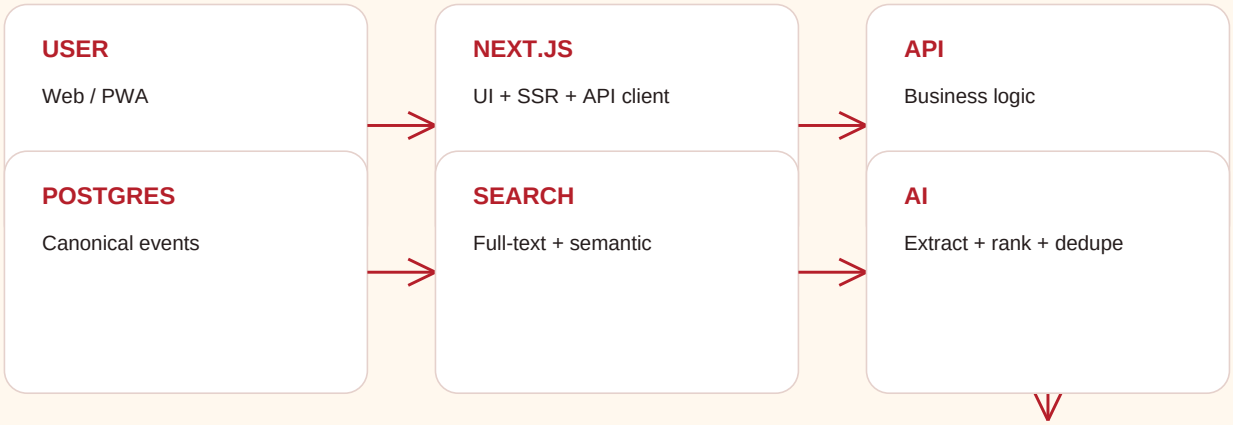
LLM extraction Recommendations Deduplication Moderation

INFRA

Vercel Object storage CDN Monitoring

Architecture goal: simple MVP → scalable discovery platform

Architecture Overview



ARCHITECTURE PRINCIPLE

Keep the core event model relational and deterministic. AI should enrich the data, not become the source of truth. Every event should retain its original source and organizer reference.

Frontend Stack

Next.js

TypeScript

Tailwind CSS

shadcn/ui

PWA

React Query / TanStack Query

Zod

React Hook Form

Fast organizer submission and profile forms

Frontend Application Structure

HOME

/

EXPLORE

/explore

EVENT

/event/[slug]

SAVED

/saved

SUBMIT

/submit

PROFILE

/profile

ADMIN

/admin

COMPONENTS

EventCard EventGrid FilterSheet MapPreview SaveButton ShareSheet

DESIGN SYSTEM

Cream background Red primary Editorial type 8px spacing rhythm Responsive cards

Backend Stack

Node.js + TypeScript

Fastify / NestJS

PostgreSQL

Supabase

Redis

BullMQ

REST API

Simple public/private contract for the web app and future mobile apps.

Database Design

EVENT

id, title, dates, price, category, status

VENUE

id, name, address, lat/lng

ORGANIZER

id, name, handles, verification

SOURCE

id, URL, platform, captured_at

USER

id, preferences

SAVE

user_id, event_id

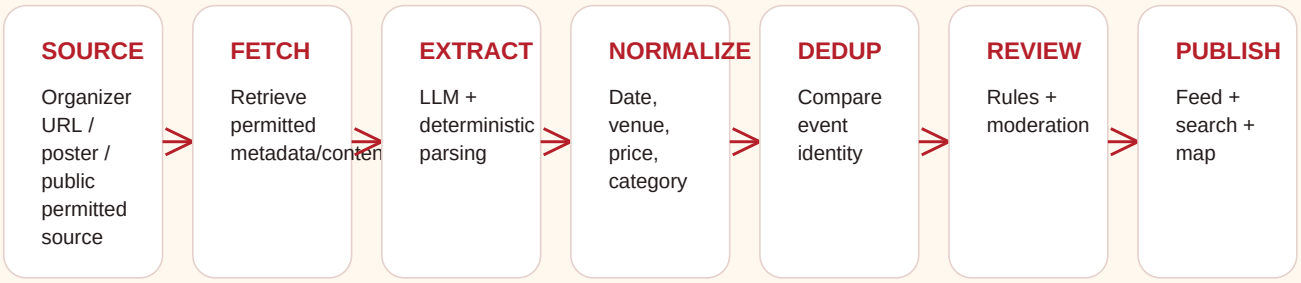
INTERACTION

type, user/event, timestamp

CRITICAL RELATIONSHIP

EVENT 1 → N SOURCE One canonical event can reference Instagram, organizer, ticketing and venue URLs without duplicating the event in the user feed.

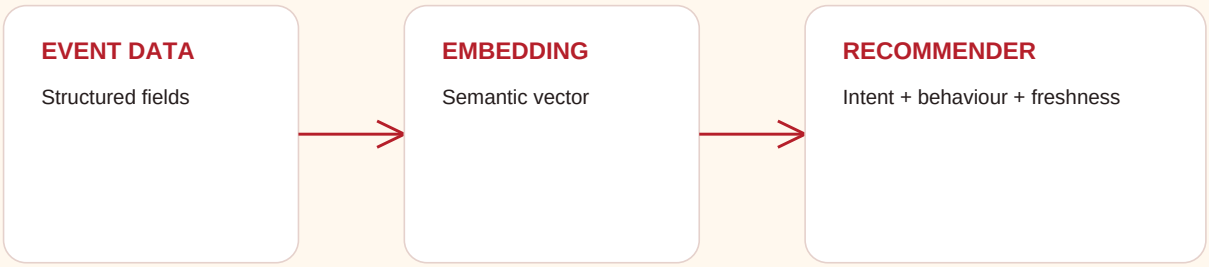
Event Ingestion Pipeline



AI SHOULD NOT BE THE FINAL AUTHORITY

Use deterministic validation for dates, required fields, source URLs and event status. AI proposes structured metadata; the system validates it before publication.

AI & Recommendation Layer



EXTRACTION

Date/time Venue Price Category Short description Source type

RANKING SIGNALS

Distance Freshness Saves Clicks Category affinity Popularity

BUILD MY WEEKEND

User prompt → structured constraints → candidate retrieval → ranking → conflict check → itinerary → original source links.

Search & Discovery

QUERY

“Comedy this Saturday”

FILTER

Date + city + category

RANK

Fresh + relevant + nearby

MVP SEARCH

Postgres full-text Trigram matching Category filters Date ranges Venue/neighbourhood

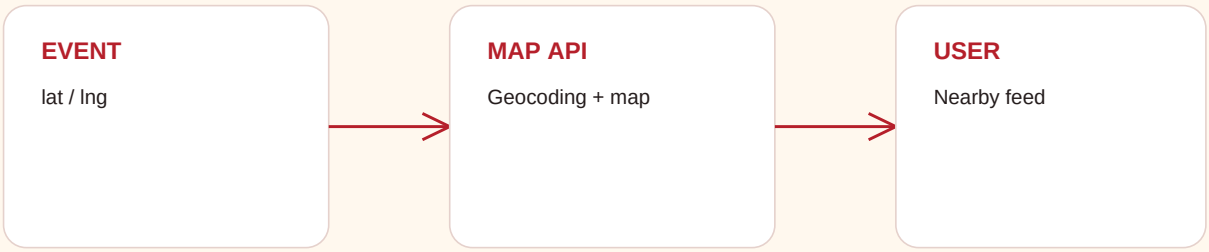
LATER

Embeddings Semantic search Natural language Personalized ranking

SEARCH EXAMPLE

“cheap date night near Park Street this Saturday” → parse budget + intent + location + date → retrieve candidate events → rank by relevance.

Maps & Location



GOOGLE MAPS / MAPBOX

Map display Directions link Geocoding Venue coordinates

LOCATION RULE

Ask for permission. Store coarse location where possible. Do not expose private location. Distance ranking should be transparent.

Authentication & Security

Supabase Auth

Row Level Security

RBAC

Rate limiting

Validation

Secrets

Audit logs

Takedown

Source removal / correction workflow

Infrastructure & Deployment

GitHub

Source control + PR review

Vercel

Frontend + serverless deployment

Supabase

Database + auth + storage

Cloudflare

DNS / CDN / security layer

Sentry

Frontend/backend error monitoring

PostHog

Product analytics

Cron / queues

Expiry + ingestion + background jobs

ENVIRONMENTS

Local → Preview → Production. Every PR gets a preview deployment. Database migrations are versioned. Production secrets are never committed.

India Market Reference Apps

BookMyShow

Events, plays, sports, activities + booking Strong inventory, filters and ticketing ■cite■turn0search3■

District by Zomato

Movies, events, dining, activities, comedy Strong “everything going out” positioning and bookable experiences

Insider

Live events / experiences Useful benchmark for event discovery and ticket-led supply

AllEvents

Large event discovery directory Benchmark for broad aggregation and city/event coverage

Meetup

Community-led events Benchmark for groups, interests and recurring communities

HighApe

Local events / experiences Benchmark for experience-led local discovery.

What We Borrow vs What We Avoid

FROM BOOKMYSHOW

Borrow: category filters, event detail structure, ticket/source clarity. Avoid: becoming only a transaction layer.

FROM DISTRICT

Borrow: going-out mindset, activities + dining + events. Avoid: requiring every experience to be bookable.

FROM INSTAGRAM

Borrow: visual feed, creator attribution, saves/shares. Avoid: unstructured information.

FROM ALLEVENTS

Borrow: breadth of event coverage. Avoid: low-quality/noisy inventory.

OUR DIFFERENTIATOR

Kolkata-first freshness + cross-platform aggregation + social discovery + “Build My Weekend” intent planning.

Recommended Final Stack

Frontend

Backend

Database

Auth

Storage

Search

AI

Maps

Jobs

Deploy

Analytics

Monitoring

Repo

GitHub + CI/CD

Build Order

PHASE 1	**Foundation** Repo, Next.js, design system, database schema, auth, event CRUD
PHASE 2	**Discovery** Home feed, categories, filters, event pages, search, save/share
PHASE 3	**Supply** Organizer submission, moderation, source model, expiry and dedup
PHASE 4	**Intelligence** AI extraction, ranking, semantic search, Build My Weekend
PHASE 5	**Growth** Organizer analytics, referral links, personalization, SEO
PHASE 6	**Scale** Queues, caching, mobile app, additional cities

ENGINEERING RULE

Do not over-engineer the first release. Start with a relational event database, strong moderation and excellent UX. Add AI, vectors, queues and native apps only when actual product usage demands them.